

INSuRE Fall 2022

Midterm Report - October 21, 2022

**Making Signals Naturally Difficult to Classify with Supervised Learning
Algorithms via Model-Agnostic Adversarial Perturbations**

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Team 1

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1 Project Summary

1.1 Project Description

Till now the attacks on classifiers were mostly through changing the data that is trained to get the wrong classifier which is the Data Poisoning or change the input during the testing, which is called the Data Evasion, also known to be the black box, grey box and the white box attacks respectively. Measures have been taken and models were evolved to overcome and identify the attackers. But zero query attacks are truly a stealthy attack. The attacker won't have any access to the victim system. The attacker can implement an attack plan once by perturbing multiple objects in each scene and submitting the perturbed input.

1.2 Estimated Project Budget

As our team does not anticipate any expenses, we do not have any project budget.

1.3 Timeline of Milestones

Key Deliverables	Due Date / Expected Completion Date
Project Proposal	September 9, 2022
Phase 1: Get a basic understanding of Adversarial Perturbations and fundamentals	September 14, 2022
Final Midterm Presentation	October 21, 2022
Phase 2: Know how to create a substitute model and how to craft adversarial samples	October 21, 2022
Phase 3: Attack the target model with these adversarial samples	November 11, 2022
Phase 4: Once we validate our attack model, then document this study and publish using some publication framework.	December 2, 2022
Final Report Presentation	December 2, 2022
Final Report	December 4, 2022

1.4 Specific Aims

We train a local model to substitute for the target DNN, using inputs synthetically generated by an adversary and labeled by the target DNN. We use the local substitute to craft adversarial examples and find that they are misclassified by the targeted DNN.

To perform a real-world and properly blinded evaluation, we attack against an unknown off-the-shelf DeepSig brand Omnisig classifier trained and supplied by Lincoln Lab. Attacks may be developed with the aid of a range of validation classifiers created by the students.

Our main aim of the project is to verify the effectiveness of the target model when the adversarial examples are passed through the target model.

2 Executive Summary

Adversarial Perturbations is a form of attack in which bad actors attempt to manipulate machine learning models that detect attacks and anomalies by “poisoning” the data they train on. These Perturbations give a understanding about the model to the attackers so that they can change the data to make the classifiers go wrong. Black box attacks, White box attacks and grey box attacks mainly target on the data to make the classifier go wrong. But the no box attacks target the meta data. We think that with experimentation and research; we could attempt to test an existing open-source dataset and model against existing no box attack or query-based attacks and to help inform the maintainers of the dangers that such techniques may face and how to mitigate them.

3 Previous Work

3.1 Evaluating adversarial evasion attacks in the context of wireless communication

Recent advancements in radio frequency machine learning (RFML) have demonstrated the use of raw in-phase and quadrature (IQ) samples for multiple spectrum sensing tasks. Yet, deep learning techniques have been shown, in other applications, to be vulnerable to adversarial machine learning (ML) techniques, which seek to craft small perturbations that are added to the input to cause a misclassification. The current work differentiates the threats that adversarial ML poses to RFML systems based on where the attack is executed from: direct access to classifier input, synchronously transmitted over the air (OTA), or asynchronously transmitted from a separate device.

3.2 Practical black-box attacks against machine learning

Machine learning (ML) models, e.g., deep neural networks (DNNs), are vulnerable to adversarial examples. All existing adversarial example attacks require knowledge of either the model internals or its training data. In this paper they introduced the first practical demonstration of an attacker controlling a remotely hosted DNN with no such knowledge.

3.3 Explaining and Harnessing Adversarial Examples

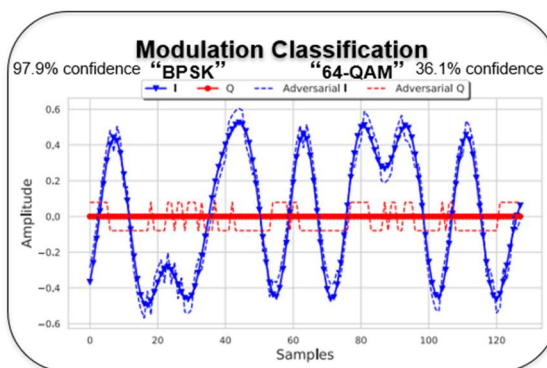
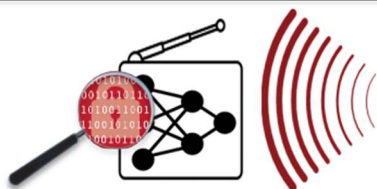
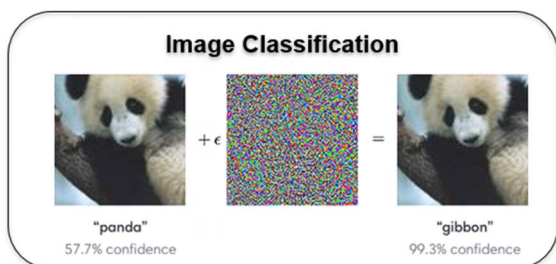
Several machine learning models, including neural networks, consistently mis- classify adversarial examples. Early attempts at explaining this phenomenon focused on nonlinearity and overfitting. This paper argues instead that the primary cause of neural networks’ vulnerability to adversarial perturbation is their linear nature. This explanation is supported by new quantitative results while giving the first explanation of the most intriguing fact about them.

3.4 Query-efficient hard-label black-box attack: An optimization-based approach

This paper studies the problem of attacking a machine learning model in the hard-label black-box setting. This is a very challenging problem since the direct extension of state-of-the-art white-box attacks to the hard-label black-box setting will require minimizing a non-continuous step function, which is combinatorial and cannot be solved by a gradient-based optimizer. The only current approach is based on random walk on the boundary, which requires lots of queries and lacks convergence guarantees. This work proposes a novel way to formulate the hard-label black-box attack as a real-valued optimization problem which is usually continuous and can be solved by any zeroth order optimization algorithm.

4 Motivation

Adversarial Perturbations is a form of attack in which bad actors attempt to manipulate machine learning models that detect attacks and anomalies by “poisoning” the data they train on. These Perturbations give a understanding about the model to the attackers so that they can change the data to make the classifiers go wrong. Black box attacks, White box attacks and grey box attacks mainly target on the data to make the classifier go wrong. But the no box attacks target the meta data. We think that with experimentation and research; we could attempt to test an existing open-source dataset and model against existing no box attack or query-based attacks and to help inform the maintainers of the dangers that such techniques may face and how to mitigate them.



- Machine learning fragility seen in computer vision also exists in communication systems
- Gathering information needed to craft attacks from passive receivers remains challenging



Unclassified//FOUO

Cybersecurity Directorate – Top Challenges FY22
Unclassified Version – 7 March 2022

Primary:

Priority: 1

PoC: Cathie Cope, Neal Ziring

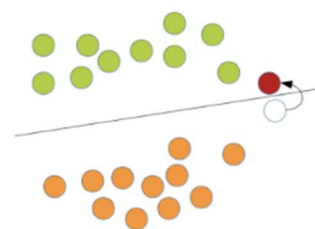
Title: (U) AI/ML in Cyber Defense

Description: (U) Two parts:

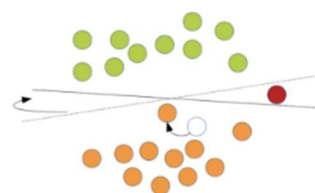
1. Detection of, prevention of, and response to malicious use of AI in cyber-attacks and identifying next-generation attack surfaces and infrastructure. How can we find and characterize and defeat adversaries when they employ AI/ML in their malicious cyber operations.
2. Use of AI/ML to improve, empower, accelerate defensive operations. Use of AI/ML to improve defensive workflow, or to enhance responses to attacks.



An increasing use of machine learning at the physical layer by DoD contractors presents a new attack surface

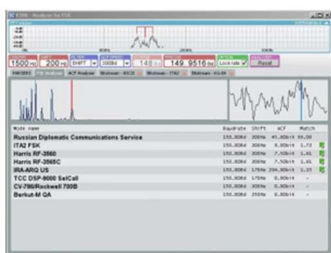


Classical adversarial attack:
directly modifying the testing sample



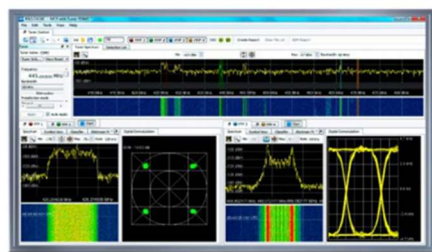
Data poisoning:
modifying training samples intelligently

Impact: Commercial Trends



Comint
consulting
Krypto500
software

Deepwaves's
AIR-T hardware

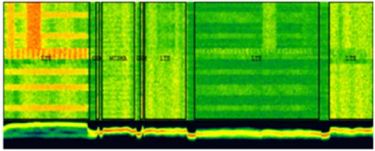
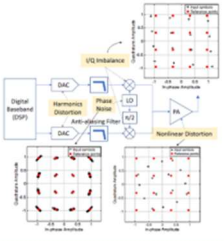
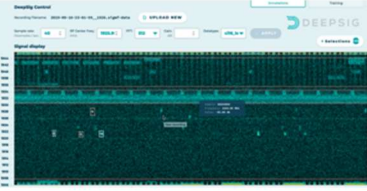


Rhode &
Schwarz's
CA100
software

Deepsig's
Omnisig
software



Impact: Academic Trends

Modulation/protocol	Emitter Identification	Detection/Coding/Channel Estimation
		
Physical layer data used: To be perturbed to deny service of classifiers		
IQ (moments, cumulants)	Impulse response, carrier phase, IQ	Spectrograms/waterfalls, channel state information, IQ

5 Research Project Plan and Approach

5.1 Research Project Phases

We plan to complete our research project in four phases. First, we will get a basic understanding of Adversarial Perturbations and fundamentals. Then we will understand about how to create a substitute model and how to craft adversarial samples. Then we try to attack the target model with these adversarial samples to misclassify and to reduce the output integrity.

- Phase 1: Get a basic understanding of Adversarial Perturbations and fundamentals
- Phase 2: Know how to create a substitute model and how to craft adversarial samples
- Phase 3: Attack the target model with these adversarial samples
- Phase 4: Once we validate our attack model, then document this study and publish using some publication framework.

5.2 Resources Required

At this time, we do not anticipate the use of any resources apart from a collection of research papers.

6 Current Progress

6.1 Explaining Adversarial Examples:

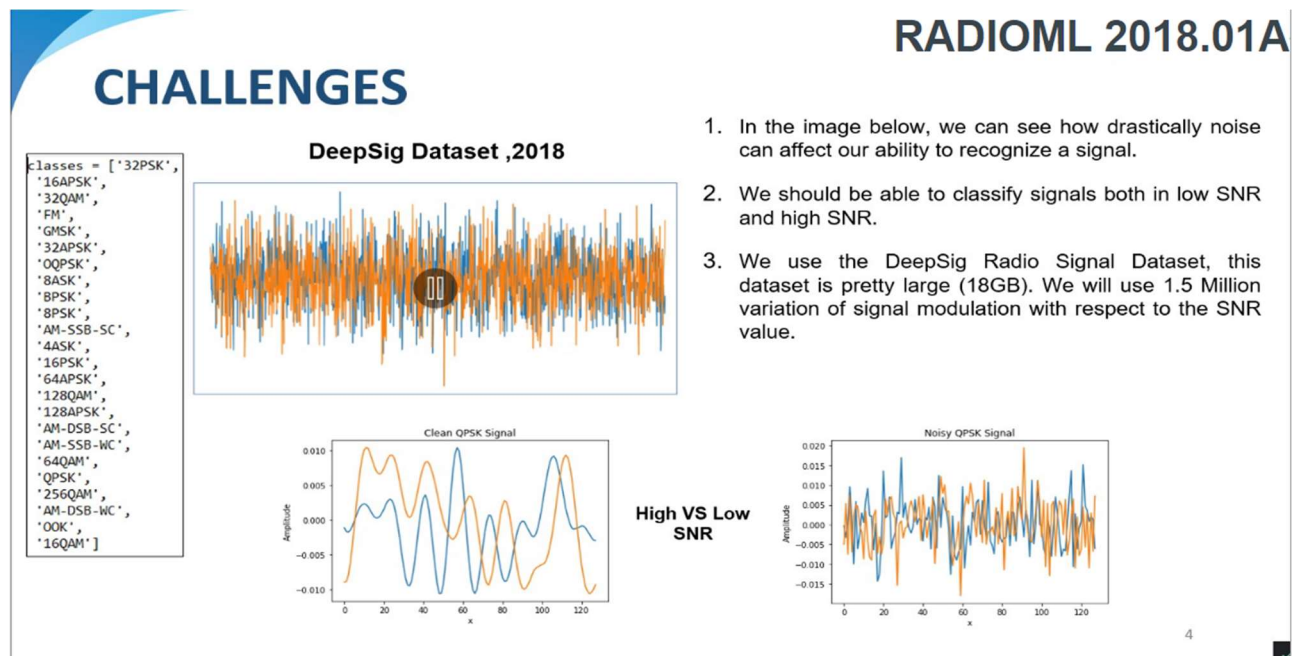
- $\tilde{x} = x + \eta$ where x is the input, $\tilde{x}$ is the adversarial input, η is the perturbation
- Expect same class prediction for x and $\tilde{x}$ if $\|\eta\| < \epsilon$, where ϵ is small enough to be discarded by the sensor or data storage apparatus associated with our problem.
- $w \cdot \tilde{x} = w \cdot x + w \cdot \eta$ w = weight vector, activation to grow by $w \cdot \eta$
- If w has n dimensions and the average magnitude of an element of the weight vector is m , then the activation will grow by mn
- Shows that a simple linear model can have adversarial examples if its input has sufficient dimensionality

- Neural networks linear behavior suggests that cheap, analytical perturbations of a linear model should also damage neural networks.
- Adversarial example generated for one model is often misclassified by other models
- It is seen that multiple classifiers assign the same class to adversarial examples.
- Adversarial examples occur in contiguous regions of the 1-D subspace defined by the fast gradient sign method. Different models learn similar functions when trained to perform the same task.
- The direction of perturbation, rather than the specific point in space, matters most.
- Because it is the direction that matters most, adversarial perturbations generalize across different clean examples.

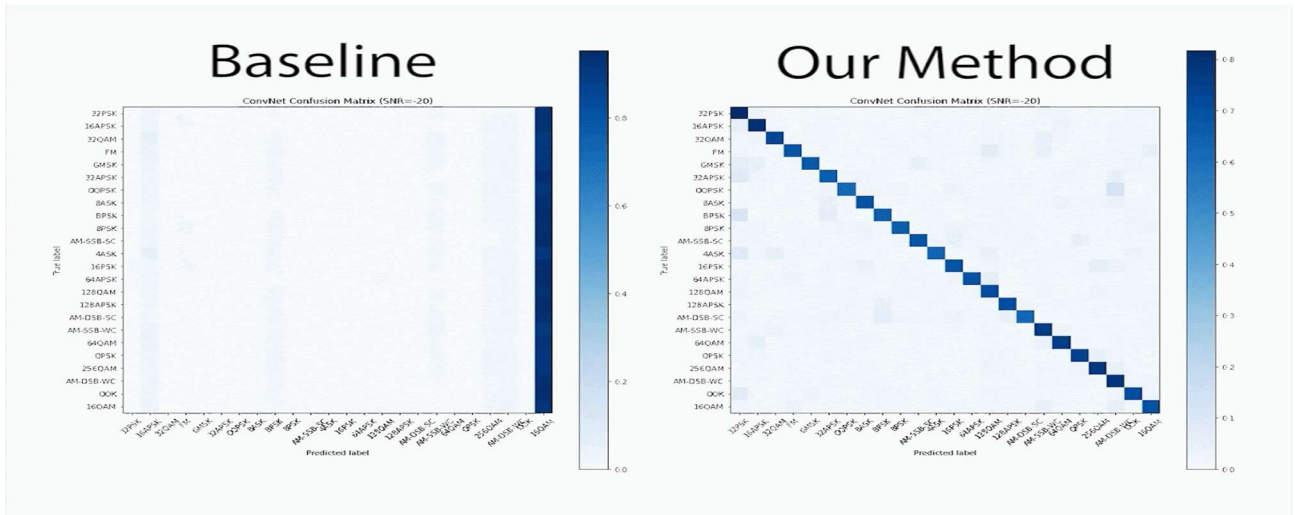
6.2 Researched on different attack techniques:

- L-BFGS In target problem, L-BFGS Attack calculated approximate values of adversarial examples by line-searching with $c > 0$ We repeatedly solve this optimization problem for multiple values of c , adaptively updating c
- FGSM FGSM attack is a white-box attack with the goal of misclassification The fast gradient sign method uses the gradient of the loss function to determine in which direction the pixel's intensity should be changed (whether it should be increased or decreased) to minimize the loss function; then, it shifts all pixels simultaneously Difference Between them:
- FGSM is designed primarily to be fast instead of producing very close adversarial examples

6.3 Dataset Extraction and Analysis:



6.4 Model Analysis:



7 Deliverables

The following due dates for the deliverables are in accordance with the INSuRE Hub program's timeline.

Key Deliverables	Due Date / Expected Completion Date
Project Proposal	September 9, 2022
Midterm Presentation Draft	October 14, 2022
Final Midterm Presentation	October 21, 2022
Final Report Presentation	December 2, 2022

8 Anticipated Issues and Challenges

As most of us are beginners in the field of ML, tackling big data sets and understanding the models would be a hurdle initially. We will try to learn different ML models and how the data set parameters are used to increase the accuracy of the model.

The concept of Adversarial Perturbations is relatively new. So, we need to get an in-depth knowledge of how it's done and what it can do to the accuracy of the model. We will try to learn how to create a substitute model and reduce the targets model accuracy. With this we can try to find out the parameters which can be easily targeted.

We use the DeepSig Radio Signal Dataset, this dataset is large (21GB). We will use 1.5 million variations of signals modulation with respect to the SNR values. Tried on multiple systems to run the data set as it is 21 GB large.

9 Investigator Biographies

Raviteja Avuthu holds a bachelors and master's degree in Computer Science and is doing his second masters at The University of Texas at Dallas, graduating in December 2023 with a focus in Data Science. He has experience working in IBM as a Software Developer. He has done Machine Learning as an undergraduate course and is looking forward to gaining expertise in the field of Data Science. He also has experience in Python and R.

Raghuveer Pachipulusu Graduated with BTech in Computer Science and currently pursuing master's in computer science at University of Texas at Dallas on Data Science Track. He Has Previously worked on Machine Learning and Deep Learning Projects and Published a Technical Paper in International Conference. He Posses Web Development Skills as well as Programming skills such as Python, Java, SQL, and R. He has taken courses in Cryptography and Network Security

Saipriya Nimmagadda holds a bachelor's degree in Computer Science and is pursuing her masters at The University of Texas at Dallas, graduating in May 2023 with a focus in Data Science and cloud computing. She started her career in Cerner healthcare Pvt Ltd as a Software Engineer with 2+ years of experience. She is a certified AWS cloud practitioner who is currently working as AWS developer for office of information technology at university of Texas at Dallas. Gaining more exposure in cloud and machine learning areas is her main goal.

Srinivas Panchangam holds a bachelor's degree in Electronic and Communications Engineering and is pursuing his masters at The University of Texas at Dallas, graduating in Dec 2023 with a focus in Data Science. He started his career as a software developer in one of the biggest tech companies in India, TCS. Though being from a different bachelors background his interest towards data science never ended, so he is trying to explore the field and get expertise in that.

With a bachelor's in computer science and an associate in management, Varun Potluri is a second-year master's student in computer science specializing in cyber security. He has experience working in a variety of fields, such as health care, risk management, software development, mergers and acquisitions, cloud security using Amazon Web Services, and cyber security. During his bachelor's, Varun published 2 research papers in cyber security space with an aim to increase the security of storage application using blockchain technology. Mr. Potluri was very interested in programming, reverse engineering, malware analysis, pen testing, risk management, and cloud security throughout his security-related studies.

10 Bibliography

- [1] Flowers, Bryse, R. Michael Buehrer, and William C. Headley. "Evaluating adversarial evasion attacks in the context of wireless communications." *IEEE Transactions on Information Forensics and Security* 15 (2019): 1102-1113.
- [2] Practical black-box attacks against machine learning.
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- [4] Query-efficient hard-label black-box attack: An optimization-based approach